RAVALI YERRAPOTHU

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Summary

- Experienced MachineLearning Engineer and a programming expert with industry experience of 6 years encompassing production level software development and prototyping and experimenting new techniques.
- Experienced Machine Learning Engineer in scientific implementation setting.
- Academic exposure to advanced Mathematics and Computer Science techniques.
- Interested in design and implementation of novel Machine Learning algorithms.
- Worked in Small team environments and Corporate infrastructure.

Expertise

Predictive Modeling, Multivariate Classification, Clustering & Regression, Statistical analysis, Time Series Analysis, Feature Engineering and Feature Reduction Reduction, Image Processing, Analysis of Large Datasets, Synthetic Data, Visual Dashboards.

Technical Skills

Languages:	C, CPP, Java, Python, R, SQL, Processing 3.0.
Open source:	OpenCV, LibSVM, numpy, pandas, scikit, Jupyter Notebook, Tableau, tensorflow AWS Sagemaker
Database:	RDBMS, Postgres, MongoDB.
Areas of Interest:	Machine Learning, Data Visualization, Computer Vision, Algorithms & Data structures.
SDLC Tools & Methodologies:	Agile Development, Git, SVN, JIRA

Education

Masters(CS) from *UNIVERSITY OF SOUTH FLORIDA*, FL, USA. ***(3.70 GPA)***. May 2017
Bachelors(CS) from *JNTUH COLLEGE OF ENGINEERING,* Hyderabad, India ***(3.65 GPA)*** May 2013

Professional Experience

Polaris Consulting Services - Consultant Technology , ***(Jan' 20 to present)***

Google Ads Dev Rel

- Responsible for design and implementation of **Google Ads** Client libraries to interact with **Google Ads API**.
- Leading the team to ensure the new feature requests are implemented to the best standard.
- Review Design documents from Google Engineering team, coordinate with them to comprehend the feature, identify the corner cases and implement test cases in client library to verify the feature.
- Designed **PLX dashboard** interface to portray a quantitative measure of client libraries, issues and test cases written by team to Google management.
- Configured build scripts for CI CD pipeline for repository and branches.
- Perform code reviews for self, and team.
- Communicate with Google Stakeholder, and Polaris Management regarding the status and roadblocks in the project implementation.

Java 8, Junit, Shell Scripting, Linux Environment, Build Management - Kokoro, Git, Gerrit

RegInsight – *Tampa, FL*

- Part of US Regulatory Reporting and Analytics team.
- Coordinated and implemented change requests and enhancement requests for **14M Monthly**, **Quarterly** and **Annual** reports.

- Implemented changes to derive **Variance** reports, **MIN-VALUE , MAX-VALUE, DEVIATION** reports for finances in all segments.
- Capturing artifacts related to projects from Initiation to testing results.
- Perform Sanity testing for the features implemented.
- Designed and developed **POCs** to develop **comparison charts** and drive through user acceptance phase to implementation and testing phases of delivery.

Python, Tableau, Apache Spark SQL.

SIOS Technology Corp* - Software Engineer - Machine Learning.,** *Columbia, SC.* ***(Oct'17 to Aug'19)

SIOS iQ, LifeKeeper, DataKeeper

- Part of **NextGen** Machine Learning team focusing on research and development of analytics in the product.
- Solved **Deserialization** problem using R scripting to enable access to serialized content of ML algorithm results (including clusters and dynamic thresholds) generated during application run time and persisted in the database.
- Researched and implemented solution to **IOPSEstimation** component as part of ***Storage Performance Forecasting***. This included understanding storage architecture and functioning in virtual, physical and cloud environments, brainstorming ideas to identify the problem, experimenting the candidate solutions and validating them followed by development and testing in the product.
- Research and development of **Multi Agent** based **Seasonality** framework to identify and extract trends and patterns from weekly, monthly and quarterly data to provide an optimized solution to the user based on seasonal behavior.
- Implementation of **Forecasting** and **Anomaly Detection** Algorithms involving amalgamation of several complex techniques like **K- Means , Cluster summarization techniques, Monte Carlo Simulation, Behavior Profile and Behavior Segment** generation for time series data, deriving statistical boundaries and value or outcome prediction.
- Have gained expertise in practical implementation of ML techniques in various areas in the field like **synthetic data simulation**, **pattern analysis**, devising tests like **T-Test, Chi-square** test and **Model evaluation** metrics.
- Used Spring Core annotations for **Dependency Injection** Spring DI.
- Developed back end persistence layer using **Hibernate** framework and integrated **Spring DAO** for data access.
- Achieved concurrent access to functionality using concepts from **Multi-threading**.
- Accomplished enhanced modules by applying **SOLID** principles and **Design Patterns**.
- Analysis of available **Azure Analytics frameworks** to aid transition of the product to Microsoft Azure framework.
- Worked extensively on complex **SQL** queries and **PLSQL scripts** to access data in necessary format.

(R, Java, J2EE, RESTful , Spring, Hibernate, JUnit, Mockito, Tomcat, postgres, Linux, OptaPlanner, Apache KMeansByMadlib)

***Communication Concepts Integrations* - Research Intern.,** *Tampa, FL* **(Dec '16 to May'17)**

VTOS(Vision Track Occupancy System)

- Implemented an Image-rectification technique **MLESAC** to rectify camera movements as part of camera calibration.
- Research and development of state-of-the-art Stereo matching algorithms for object detection and depth calculation to generate disparity map in 3D model reconstruction of the scene.
- Used block matching algorithms and measured dissimilarity using Minimization – Propagation approach which gave better results in the application compared to **StereoBM** and **StereoSGBM** algorithms.
- Developed a classifier model using **Support Vector Machines** to classify good vs bad weather conditions. This included data extraction, image preprocessing and feature extraction, feature selection and parameter tuning to find the best hypothesis. Improved the accuracy to 87% from 73. Used **Weka** for model selection and **Java** & **LibSVM** for development.

(OpenCV, Java, C++,LibSVM, Matlab)

***IBM India Private Ltd* - Application Developer.,** *Bangalore, India* **(Feb'14 to Nov'15)**

ABSCON (AT&T's Business Contracting)

- Responsible for Analyzing, designing, developing, co-ordinating and deploying web based automated Contracting application.
- Designed and developed User Interaction screens using HTML5, CSS3, JavaScript, AJAX and JSON.
- Performed unit testing and debugging using **Junit** and **Mockito** for reported issues.
- Quickly assumed the role of SME for the application.
- Used Tortoise SVN for version control of source code.
- Worked in both Agile and Waterfall development methodologies.

HTML, CSS, JavaScript, Java/J2EE, SQL, Oracle.

Academic Projects

Machine Learning:

Study of various Machine Learning techniques in Classification, Regression, Clustering and their Mathematical background. State-of-the-art implementation practices for performance and scalability. Loss measures, Optimization practices for algorithms and evaluation metrics.

Comparison of Linear, Quadratic and Naive Bayes Discriminant.

- Predict and analyze output of Quadratic and Naive Bayes Discriminant models built using MATLAB.

MATLAB

Analysis of ANN:

- Built and analyzed single layer and multi layer perceptron with iris dataset and performed parameter tuning to understand the impact of parameters using Weka GUI.
- Compared the performance of J48 decision tree with ANN.
- Analyzed ANN on XOR data by altering momentum, learning rate and epoch.

Analysis of CNN on MNIST Dataset:

- The CNN model consisted of 2 convolutional layers each followed by 2X2 max pooling layer and fully connected layer of 512 units. Kernel used 3X3. Built using Python 2.7 and TensorFlow API.
- Obtained an accuracy of 99.2% by increasing the Epoch and featuremaps(depth) and kernelsize.

Python, Tensorflow

Streaming Based Twitter Spam Classification:

- Built an incremental classifier that classifies spam tweets based on features from user Tweet and user Account.
- Performed Data Extraction, cleansing and feature selection. Dataset consisted of 10k spam vs non-spam tweets of continuous and non–continuous periods. Built models out of Naïve Bayes, Random Forest and MOA Classifiers.
- Compared the models using Kappa Statistic and RMS, MAE, RAE, RRSE and identified statistically significant model using F-Measure.

(Java, Weka, Trend Micro Web Reputation Service)

Computer Vision:

Study of state of the art techniques in field of CV which involved research and application of advanced Mathematical techniques involving Statistics, second order derivatives (Laplacian), Fourier transformation, Object detection and edge detection techniques.

Jigsaw Puzzle Solver:

- Term Project to develop software that extracts the image puzzle pieces, identifies the Automatic features, segregates the pieces based on its shape, finds the best fit piece at each step, assembles and merges the individual pieces of a Jigsaw puzzle given in the image form. Involved complex programming techniques like back-propagation to solve the puzzles.

Motion detection and Vehicle tracking:

- Developed software to detect number of Vehicles in a video, classify them into vehicle categories Car, Truck, or bike, display the color of the vehicles and write the maximum areal frame into images.

Image Stitching:

- Software extracts and identifies similar features from given input images, calculates the warping parameters, and stitches them using the homography matrix to create a panoramic image with combined view.

(OpenCV, C++)

Secure Instant Peer-Peer Messaging Application:

- Built an end to end multiclient – server messaging application which ensures secure transmission using advanced network encryption techniques.
- Established server resources using Java Threads and ensured the state is synchronous.
- Used socket API to establish connection between client and server.
- Symmetric key cryptography technique "AES" with periodic key update was used for encryptions and decryptions.
- Implemented a secure storage of passwords using "SHA-1" hashing technique.

(Java, Encryption/Decryption API, Socket Programming API)